MATH3900

# Tutorial 3

1. Determine whether each statement is *True* or *False*. Give a short proof for a true statement, or a counterexample with verification for a false statement.

   (a) Let $f(x) = \frac{1}{2}x^T Q x$, where $Q = Q^T \succ 0$. Multiplying $f$ by a positive number leaves its condition number unchanged.

   (b) If the Hessian $H = \nabla^2 f(x)$ is invertible, the Newton direction $p$, defined by $Hp = -\nabla f(x)$, always minimizes the quadratic Taylor model at $x$.

   (c) If $H = \nabla^2 f(x) \succ 0$ and $\nabla f(x) \neq 0$, the Newton direction is a descent direction at $x$.

   (d) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, and let $p$ be a descent direction at $x$. If $\alpha > 0$ satisfies the Armijo condition with parameter $c_A \in (0, 1)$, then $\alpha/2$ satisfies the same condition.

2. Consider logistic regression with parameter $x = (x_1, x_2)^T$ and six labelled observations:

| $i$ | $a_i^T$ | $y_i$ |
|---|---|---|
| $1, 2$ | $(1, 0)$ | $+1$ |
| $3$ | $(1, 0)$ | $-1$ |
| $4, 5$ | $(0, 3)$ | $+1$ |
| $6$ | $(0, 3)$ | $-1$ |

$$F(x) = \frac{1}{6}\sum_{i=1}^{6} \log(1 + e^{-y_i a_i^T x}).$$

   Define $\sigma(t) = e^t/(1 + e^t)$ and $w(t) = \sigma(t)(1 - \sigma(t))$. You may use

$$\log(1 + e^{-t}) = \log(1 + e^t) - t, \qquad \sigma'(t) = w(t), \qquad 0 < w(t) \leq \tfrac{1}{4}.$$

   (a) Write $F(x)$ explicitly in terms of $x_1, x_2$, and simplify it using the identity above. Find $\nabla F(x)$ and $\nabla^2 F(x)$.

   (b) Find the smallest global gradient Lipschitz constant $L$ with respect to the Euclidean norm. Explain why a smaller constant cannot work.

   (c) Starting from $x^{(0)} = (0, 0)^T$, compute one gradient descent step with step size $1/L$, and one full Newton step. For Newton, solve

$$\nabla^2 F(x^{(0)})p = -\nabla F(x^{(0)}), \qquad x_{\mathrm{N}}^{(1)} = x^{(0)} + p.$$

   Compare the two new points with the unique minimizer $x^\star = (\log 2, \frac{1}{3}\log 2)^T$. Which is closer? Does Newton reach $x^\star$ in one step? You may use $\log 2 \approx 0.6931$.

3. Let

$$f(t) = \frac{t^3}{3} - t, \qquad t > 0.$$

(a) Compute $f'(t)$ and $f''(t)$, and find the minimizer $t^\star$. At $t^{(0)} = 2$, write the quadratic Taylor model in the displacement $p$.

(b) Derive the full Newton update for any $t^{(k)} > 0$. Starting from $t^{(0)} = 2$, compute $t^{(1)}$ and $t^{(2)}$.

(c) On $D = [1/2, 3/2]$, show that $f''$ is 2-Lipschitz and $f''(t) \geq 1$. For $e_k = t^{(k)} - 1$, you may use

$$e_{k+1} = \frac{e_k^2}{2t^{(k)}}.$$

Show that $|e_k| \leq 1/2$ implies $|e_{k+1}| \leq |e_k|^2 \leq 1/4$. Explain how this gives local quadratic convergence.

4. Let

$$f(x) = \tfrac{1}{2}(x_1^2 + 3x_2^2), \qquad x = (3, 1)^T, \qquad g = \nabla f(x), \qquad p = -g.$$

(a) Compute $g$, $g^T p$, and $\phi(\alpha) = f(x + \alpha p)$. Find the exact line-search step over $\alpha \geq 0$, the new point, and its objective value.

(b) With $c_A = 1/4$, solve the Armijo inequality

$$f(x + \alpha p) \leq f(x) + c_A \alpha g^T p$$

to find all acceptable positive step sizes.

(c) Check $\alpha = 1/4$ and $\alpha = 5/6$. For each, give the objective value and the Armijo bound. Use these two trials to distinguish an exact line-search step, a step satisfying Armijo, and a step that merely decreases $f$.